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BRST–DS Compatibility on $\widehat{\mathfrak{osp}}(1|2)$

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Abstract: We study the compatibility between BRST quantization and Drinfeld–Sokolov reduction for the affine superalgebra $\widehat{\mathfrak{osp}}(1|2)$. Starting from an $\widehat{\mathfrak{sl}}2$ current algebra coupled to a complex fermion, we construct the improved Cartan current $H(z) = J^0(z) + 2 :!\psi\psi^*!: (z)$, and formulate the BRST complex with both bosonic and fermionic ghosts. We then consider the second odd operator in the resulting extended BRST structure and show that its correction term Δ is naturally characterized by cohomological equations rather than being fixed only by representation theory. In particular, the compatibility conditions take the form $[Q_{\mathrm{BRST}}, \Delta] = H_{\mathrm{osp}} - L_0, \qquad F = [b_0, \Delta] + \frac{1}{2}[\Delta, \Delta]$.

We further interpret these equations as components of a flat superconnection on the graded free-field bundle. The corresponding oscillator character can be expressed in terms of theta functions and naturally defines a determinant-line contribution associated with the boson–fermion polarization. After BRST reduction, the surviving states organize into the $N = 1$ super-Virasoro sector. This gives a concrete realization of the relation $\widehat{\mathfrak{osp}}(1|2) \longrightarrow$ BRST/DS reduction $\longrightarrow W(\mathfrak{osp}(1|2)) \simeq N = 1$ super-Virasoro, while making explicit the additional cohomological and bundle-theoretic structure underlying the reduction.

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We set

$$H(z) = J^0(z) + 2 : \psi\psi^* : (z) \tag{0.1}$$

Under this construction, we try check does the BRST and DS reduction are compatible or not.

1 $\mathfrak{osp}(1|2)$

The Lie superalgebra $\mathfrak{osp}(1|2)$ is generated by the even elements E, F, H and odd elements Q_+, Q_-, with relations

$$[H, E] = 2E, \quad [H, F] = -2F, \quad [E, F] = H \quad [H, Q_\pm] = \pm Q_\pm \tag{1.1}$$

The odd generators satisfy

$$\{Q_+, Q_+\} = 2E \quad \{Q_-, Q_-\} = 2F, \quad \{Q_+, Q_-\} = H \tag{1.2}$$

The odd component of $\mathfrak{osp}(1|2)$ is the fundamental representation of the even subalgebra:

$$\mathfrak{osp}(1|2)_{\bar{1}} \simeq \mathbf{2}_{\mathfrak{sl}_2} \tag{1.3}$$

The complex algebra admits the split real form

$$\mathfrak{osp}(1|2, \mathbb{R}) \subset \mathfrak{osp}(1|2, \mathbb{C}) \tag{1.4}$$

with involution

$$E^\dagger = F \quad H^\dagger = H \quad Q_+^\dagger = Q_- \tag{1.5}$$

This real structure extends naturally to the free-field realization by requiring

$$\psi^\dagger = \psi^* \tag{1.6}$$

The split real form is important because the noncompact directions in the Cartan decomposition

$$\mathfrak{g}_\mathbb{R} = \mathfrak{t} \oplus \mathfrak{p} \tag{1.7}$$

behave as odd directions after Clifford quantization. The algebra $\mathfrak{osp}(1|2)$ is the minimal superalgebra encoding this boson-fermion correspondence.

2 Free field realization

We now construct the affine realization used in the BRST reduction. The even sector is generated by an affine $\widehat{\mathfrak{sl}}_2$ current algebra,

$$J^+(z), J^0(z), J^-(z) \tag{2.1}$$

with operator products

$$J^0(z)J^0(w) \sim \frac{2k}{(z-w)^2} \tag{2.2}$$

$$J^0(z)J^\pm(w) \sim \pm\frac{2J^\pm(w)}{z-w} \tag{2.3}$$

$$J^+(z)J^-(w) \sim \frac{k}{(z-w)^2} + \frac{J^0(w)}{z-w} \tag{2.4}$$

The fermionic sector consists of a complex fermion pair

$$\psi(z), \psi^*(z) \tag{2.5}$$

with

$$\psi(z)\psi^*(w) \sim \frac{1}{z-w} \tag{2.6}$$

The corresponding Fock space is

$$\mathcal{H}_\psi = \Lambda(\psi_{-r}, \psi^*_{-r}) \tag{2.7}$$

thus we define

$$H(z) = J^0(z) + 2 : \psi\psi^* : (z) \tag{2.8}$$

The current $H(z)$ gives the correct Cartan action on the fermionic sector and realizes the affine Cartan generator of $\widehat{\mathfrak{osp}}(1|2)$. It is easy to check by OPE of operators. Hence the improved current shifts the $\mathfrak{sl}_2$ Cartan charge by the spinorial contribution. This gives the required weights of the odd generators $Q_\pm$.

3 BRST

The BRST resolution introduces ghosts (b, c) for the bosonic generator E and (β, γ) for the fermionic generator Q_+.

The total state space is

$$\mathcal{H}_{\text{tot}} = \mathcal{H}_B \otimes \mathcal{H}_\psi \otimes \mathcal{H}_{bc} \otimes \mathcal{H}_{\beta\gamma} \tag{3.1}$$

The $\mathfrak{osp}(1|2)$ structure requires a second odd operator, we define

$$Q_+ = Q_{\text{BRST}} \tag{3.2}$$

$$Q_- = b_0 + \Delta \tag{3.3}$$

where Δ contains the fermion contribution. explictly

$$Q_{\rm BRST} = \oint dz(c_a J^a - \frac{1}{2} f^a_{bc} c_a c^b c^c + \gamma_i \Psi^i + \cdots) \tag{3.4}$$

and

$$\{Q_{\rm BRST}, b_0\} = L_0 \quad \{Q_{\rm BRST}, Q_{\rm BRST}\} = 0 \tag{3.5}$$

Since the BRST quantization need satisfied the quantum master equation, we define

$$\mathcal{Q} = Q_+ + \vartheta Q_- \tag{3.6}$$

where

$$\vartheta^2 = 0 \tag{3.7}$$

thus

$$\frac{1}{2}[\mathcal{Q}, \mathcal{Q}] = \frac{1}{2}[Q_+, Q_+] + \vartheta[Q_+, Q_-] + \frac{1}{2}\vartheta^2[Q_-, Q_-] = 0 \tag{3.8}$$

since $Q_+^2 = 0$, thus we need to consider the other two terms. We expand

$$[Q_+, Q_-] = [Q_+, b_0] + [Q_+, \Delta] = L_0 + [Q_+, \Delta] \tag{3.9}$$

thus

$$H = L_0 + [Q_+, \Delta] \tag{3.10}$$

thus the Δ is determinate by

$$[Q_{\rm BRST}, \Delta] = J^0 + 2 : \psi\psi^* : + H_{\rm ghost} - L_0 = H_{\rm osp} - L_0 \tag{3.11}$$

also under our assumption

$$F = [b_0, \Delta] + \frac{1}{2}[\Delta, \Delta] \tag{3.12}$$

thus the Δ is not choosen by representation theory, but it is the solution of the cohomological equation

$$[Q_{\rm BRST}, \Delta] = H_{\rm osp} - L_0 \tag{3.13}$$

$$F = [b_0, \Delta] + \frac{1}{2}[\Delta, \Delta] \tag{3.14}$$

4 On line bundle

The unreduced free-field system has character

$$Z(q, z) = \prod_{n \geq 1} \frac{(1 + zq^{n-\frac{1}{2}})(1 + z^{-1}q^{n-\frac{1}{2}})}{(1 - zq^n)(1 - z^{-1}q^n)} \tag{4.1}$$

Introducing

$$q = e^{2\pi i \tau}, \qquad z = e^{2\pi i u}, \tag{4.2}$$

the product may be rewritten in terms of theta functions as

$$Z(q,z) = iz^{-1/2}q^{1/8}\frac{\vartheta_3(u|\tau)}{\vartheta_1(u|\tau)}. \tag{4.3}$$

More intrinsically, the partition function has the form

$$Z = \mathrm{sdet}^{-1/2}(D), \tag{4.4}$$

where D denotes the kinetic operator on the graded oscillator complex. This defines a determinant line

$$\mathcal{L}_{\mathrm{det}} = \det(D_{\bar{0}}) \otimes \det(D_{\bar{1}})^{-1}. \tag{4.5}$$

actually, here we introduced this character is already enhanced the supersymmetry from $\mathcal{N} = 2$ to $\mathcal{N} = 1$. We would see that this realization on the vector bundles already given it a $U(1)$ R-symmetry.

The BRST complex contains contractible pairs

$$(E, b, c), \quad (Q_+, \beta, \gamma) \tag{4.6}$$

Each quartet contributes the ratio

$$\frac{\det_{\mathrm{fermion}}}{\det_{\mathrm{boson}}} = 1 \tag{4.7}$$

The surviving states are generated by

$$L_{-n}, \quad n \geq 2, \qquad\qquad G_{-r}, \quad r = \frac{3}{2}, \frac{5}{2}, \ldots \tag{4.8}$$

G are already indicate this representation is in NS sector. the reduced character is

$$\chi_W(q) = q^{-c/24}\frac{\prod_{n\geq 1}(1+q^{n+\frac{1}{2}})}{\prod_{n\geq 2}(1-q^n)} \tag{4.9}$$

The free-field construction naturally defines a graded vector bundle.

$$\mathcal{E} = \mathcal{E}_{\bar{0}} \oplus \mathcal{E}_{\bar{1}} \tag{4.10}$$

The bosonic and fermionic sectors contribute respectively

$$S(\mathcal{E}_{\bar{0}}), \quad \Lambda(\mathcal{E}_{\bar{1}}) \tag{4.11}$$

suppose A is the connection one-form on the bundle. Then the BRST superconnection on the bundle is

$$\nabla_{\mathrm{BRST}} = d + Q_{\mathrm{BRST}} + b_0 + \Delta + A \tag{4.12}$$

the flatness condition is

$$\nabla^2_{\mathrm{BRST}} = 0 \tag{4.13}$$

where the expansion is

$$dA + A^2 + [Q_{\mathrm{BRST}}, \Delta] + [b_0, \Delta] + \frac{1}{2}[\Delta, \Delta] + \cdots = 0 \tag{4.14}$$

component equations reproduce

$$[Q_{\rm BRST}, \Delta] = H_{\rm osp} - L_0 \tag{4.15}$$

$$F = [b_0, \Delta] + \frac{1}{2}[\Delta, \Delta] \tag{4.16}$$

Let $\mathfrak{g}_{\mathbb{C}}$ be a complex reductive Lie algebra with split real form $\mathfrak{g}_{\mathbb{R}}$. The Cartan decomposition is

$$\mathfrak{g}_{\mathbb{R}} = \mathfrak{t} \oplus \mathfrak{p}. \tag{4.17}$$

The subspace $\mathfrak{p}$ provides the noncompact directions. Under Clifford quantization these directions generate spinorial degrees of freedom. This explains the natural appearance of $\mathfrak{osp}(1|2)$ as the minimal principal embedding algebra of split real forms of Lie algebra in $\mathcal{N} = 2$ supersymmetry case. It gives a natural $U(1)$ R-symmetry which enhanced it. And $q^{1/8}$ records the determinant-line contribution of the fermionic-bosonic polarization.

Thus, overall structure is as follow

$$\boxed{\begin{array}{c} \widehat{\mathfrak{osp}}(1|2) \\ \downarrow \\ (Q_+, Q_-) \text{ extended BRST structure in } N = 2 \\ \downarrow \\ Q_{\rm BRST}\text{-cohomology} \\ \downarrow \\ W_k(\mathfrak{osp}(1|2)) = N = 1 \text{ super}-\text{Virasoro} \end{array}}$$